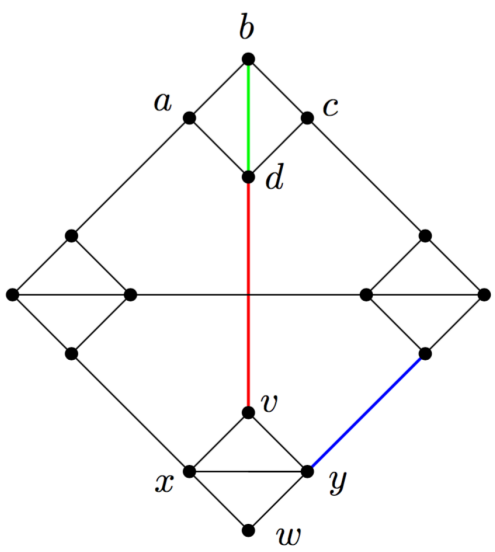
## Your grade: 85.71%

Your latest: **85.71%** • Your highest: **85.71%** • To pass you need at least 60%. We keep your highest score.

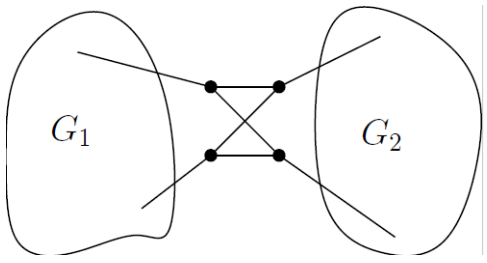
Next item

**1.** Consider the graph below. Which of the followings statements are true? 1 / 1 point




Nice work

**2.** Look at the graph $G$ below, in which two (unknown to us) graphs $G_1$ and $G_2$ are connected by some "crossover gadget'' of four vertices. Which of the following statements are true for all such $G$, i.e., regardless of $G_1, G_2$? 0 / 1 point




Try again

**3.** We consider the *Hamming cube*, which is a graph on $2^n$ vertices, defined as follows. Its vertex set is $\{0,1\}^n$, the set of binary strings of length $n$. Two vertices $u, v$ are connected if they differ in exactly one component. If you imagine $\{0,1\}^n$ as a (finite) subset of the space $\mathbb{R}^n$, then $u, v$ are connected by an edge if and only if their Euclidean distance is $1$. We denote this graph by $H_n$. How many Hamiltonian cycles does $H_3$ have? 1 / 1 point

Nice work

**4.** Let $f(G)$ be the minimum number of edges you have to remove from $G$ to destroy every Hamiltonian cycle. For example, $f(C_n) = 1$ since $C_n$ has a Hamiltonian cycle (it is itself a cycle) and destroying one edge makes it non-Hamiltonian. 1 / 1 point

**What is $f(H_4)$?**

Nice work

**5.** Consider $f(G)$ as defined above. 1 / 1 point

**What is $f(H_3)$?**

Nice work

**6.** In the peer review, you were asked to come up with an explicit formula for $f(K_n)$. That is, how many edges do you have to remove from the complete graph $K_n$ to destroy all Hamilton cycles? 1 / 1 point

In this and the following exercises, you will need this formula, but you won't have to prove it.

**What is $f(K_{50})$?**

Nice work

**7.** What is $f(K_{99})$? 1 / 1 point

Nice work